

- Description

This module manages a smart boiler temperature control system, ensuring that:

1. **The boiler temperature is continuously monitored and maintained within a safe range.**
2. **If the temperature exceeds the upper threshold, an overheat warning is triggered.**
3. **If the temperature drops below the minimum threshold, the heating system is activated.**
4. **A log of all temperature readings, warnings, and adjustments is maintained for safety tracking.**

This system helps prevent overheating hazards while ensuring efficient temperature regulation in industrial applications.

- Specification

module BoilerTemperatureMonitoring

var currentTemp, maxTemp, minTemp: real

var tempLog: sequence of string

process updateTemperature(changeAmount: real)

functional scenario form:

currentTemp <= maxTemp && tempLog = tempLog $\oplus$ ("Temperature Updated: " + currentTemp + "°C"), Temperature updated successfully.

||

currentTemp > maxTemp && tempLog = tempLog $\oplus$ ("Overheat Warning: " + currentTemp + "°C"), Overheat warning triggered.

||

currentTemp < minTemp && tempLog = tempLog $\oplus$ ("Low Temperature Alert: " + currentTemp + "°C, Heating Activated."), Low temperature alert triggered, heating activated.

end process

end module

- Java Code

```
package code;
import java.util.*;
class Boiler {
    private double maxTemp;
    private double minTemp;
    private double currentTemp;
    private List<String> tempLog;
    public Boiler(double maxTemp, double minTemp, double initialTemp) {
        this.maxTemp = maxTemp;
        this.minTemp = minTemp;
        this.currentTemp = initialTemp;
    }
}
```

```

        this.tempLog = new ArrayList<>();
    }
    public void updateTemperature(double changeAmount) {
        currentTemp += changeAmount;
        String message = String.format("Current Temperature: %.1f°C", c
urrentTemp);
        if (currentTemp > maxTemp) {
            message += " | WARNING: Overheat Detected!";
        } else if (currentTemp < minTemp) {
            message += " | ALERT: Temperature Too Low! Activating Heati
ng System.";
        }
        System.out.println(message);
        tempLog.add(message);
    }
    public void printTempLog() {
        System.out.println("\nTemperature Log:");
        for (String log : tempLog) {
            System.out.println(log);
        }
    }
}

public class c29_BoilerTemperatureSystem {
    public static void main(String[] args) {
        Boiler boiler = new Boiler(100.0, 50.0, 70.0);
        // tasecase-VT:
        //     boiler.updateTemperature(20.0);
        //     boiler.updateTemperature(-30.0);
        //     boiler.updateTemperature(-20.0);
        //     boiler.updateTemperature(50.0);
        //     boiler.updateTemperature(15.0);
        // tasecase-FT:
        //     boiler.updateTemperature(-1.7976931348614471E308);
        //     boiler.updateTemperature(2.7430620343943882E303);
        //     boiler.updateTemperature(2.5421930713382118E303);
        //     boiler.updateTemperature(1.6029092485950715E293);
        //     boiler.updateTemperature(4.103512970007259E295);

        // tasecase-Z3:
        //     boiler.updateTemperature(20);
        //     boiler.updateTemperature(30);
        //     boiler.updateTemperature(20);
        //     boiler.updateTemperature(50);
        //     boiler.updateTemperature(15);
    }
}

```

```

// tasecase-UVT:
boiler.updateTemperature(20.0);
boiler.updateTemperature(30.0);
boiler.updateTemperature(20.0);
boiler.updateTemperature(50.0);
boiler.updateTemperature(15.0);

boiler.printTempLog();
}
}

```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c29_BoilerTemperatureSystem.java	100.0 %	6	0	6
> Boiler	100.0 %	6	0	6
> c29_BoilerTemperatureSystem		0	0	0

3. Vibration testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	106	17,856	17,962
▼ src	0.6 %	106	17,856	17,962
▼ code	0.6 %	106	17,856	17,962
▼ c29_BoilerTemperatureSystem.java	97.2 %	106	3	109
> c29_BoilerTemperatureSystem	89.3 %	25	3	28
> Boiler	100.0 %	81	0	81

4. Fuzz testing branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c29_BoilerTemperatureSystem.java	66.7 %	4	2	6
> Boiler	66.7 %	4	2	6
> c29_BoilerTemperatureSystem		0	0	0

5. Fuzz testing instruction coverage: 93.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	102	17,860	17,962
▼ src	0.6 %	102	17,860	17,962
▼ code	0.6 %	102	17,860	17,962
▼ c29_BoilerTemperatureSystem.java	93.6 %	102	7	109
> Boiler	95.1 %	77	4	81
> c29_BoilerTemperatureSystem	89.3 %	25	3	28

6. Z3 branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	835	840
▼ src	0.6 %	5	835	840
▼ code	0.6 %	5	835	840
▼ c29_BoilerTemperatureSystem.java	83.3 %	5	1	6
> Boiler	83.3 %	5	1	6
> c29_BoilerTemperatureSystem		0	0	0

7. Z3 instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	103	22,295	22,398
▼ src	0.5 %	103	22,295	22,398
▼ code	0.5 %	103	22,295	22,398
▼ c29_BoilerTemperatureSystem.java	94.5 %	103	6	109
> c29_BoilerTemperatureSystem	89.3 %	25	3	28
> Boiler	96.3 %	78	3	81

8. Unimproved V-method branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	835	840
▼ src	0.6 %	5	835	840
▼ code	0.6 %	5	835	840
▼ c29_BoilerTemperatureSystem.java	83.3 %	5	1	6
> Boiler	83.3 %	5	1	6
> c29_BoilerTemperatureSystem		0	0	0

9. Unimproved V-method instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	103	22,295	22,398
▼ src	0.5 %	103	22,295	22,398
▼ code	0.5 %	103	22,295	22,398
▼ c29_BoilerTemperatureSystem.java	94.5 %	103	6	109
> c29_BoilerTemperatureSystem	89.3 %	25	3	28
> Boiler	96.3 %	78	3	81